E3 Notes: geometry, proof and conic sections

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This module has three parts: **synthetic proof** in the plane (§1–5), **analytic methods** and loci leading to the **conic sections** (§6–9), and **vectors in space** (§10–11). Draw every figure, by hand or in GeoGebra. A good figure is half a proof, but a figure is never a proof.

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1. How to write a geometric proof

Dutch exams and textbooks use a fixed layout, and it is a good one:

- **Gegeven** (given): what you may assume.
- **Te bewijzen** (to prove): the claim, precisely.
- **Bewijs** (proof): a chain of statements, each followed by its reason in brackets: a given fact, a definition, or a named theorem.

Basic facts you may use (proved in stage B):

- Vertical angles (*overstaande hoeken*) are equal. Angles on a straight line add to $180°$.
- With parallel lines, corresponding angles (*F-hoeken*) and alternate angles (*Z-hoeken*) are equal.
- The angles of a triangle add up to $180°$; an exterior angle (*buitenhoek*) equals the sum of the two non-adjacent interior angles.
- In an isosceles triangle (*gelijkbenige driehoek*) the base angles are equal, and conversely.
- Congruence criteria (*congruentiegevallen*) ZZZ, ZHZ, HZH, ZHH and ZZR (the Dutch names: Z = side, H = angle, R = right angle; ZZR is RHS).
- Similarity criteria (*gelijkvormigheid*): two equal angles (hh), or proportional sides with an equal included angle.
- A tangent (*raaklijn*) to a circle is perpendicular to the radius at the point of contact.

Strategy. Angle chasing: name one or two unknown angles α, β, express every other angle in them, and see what falls out. Look for isosceles triangles hidden in radii: **two radii always make an isosceles triangle**.

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2. Thales' theorem

Theorem (Thales). If AB is a diameter of a circle and C is another point on the circle, then $\angle ACB = 90°$.

Proof. Let M be the centre. Then $MA = MB = MC$ (radii). Triangle AMC is isosceles, so $\angle MAC = \angle MCA = \alpha$; triangle BMC is isosceles, so $\angle MBC = \angle MCB = \beta$. The angles of triangle ABC add to $180°$: $\alpha + \beta + (\alpha + \beta) = 180°$, so $\angle ACB = \alpha + \beta = 90°$. ■

Converse. If $\angle ACB = 90°$, then C lies on the circle with diameter AB.

Proof. Let M be the midpoint of AB. Reflect C in M to get D. Then $ACBD$ is a parallelogram (diagonals bisect each other) with a right angle, so it is a rectangle, and the diagonals of a rectangle are equal: $CD = AB$. Hence $MC = \frac{1}{2}CD = \frac{1}{2}AB = MA$, so C is on the circle with centre M through A and B. ■

So: **the locus of points C with $\angle ACB = 90°$ is the circle with diameter AB, without A and B.** This "Thales circle" (*cirkel van Thales*) is constantly useful: whenever you see two right angles on the same segment, you get four points on a circle.

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3. The inscribed angle theorem

Let A and B be points on a circle with centre M. The chord (*koorde*) AB divides the circle into two arcs (*bogen*). For a point C on the circle, not on arc AB itself, $\angle ACB$ is an **inscribed angle** (*omtrekshoek*) on arc AB; $\angle AMB$ (measured on the side of that arc) is the **central angle** (*middelpuntshoek*).

Theorem (inscribed angle). The inscribed angle is half the central angle on the same arc: $\angle ACB = \frac{1}{2}\angle AMB$. In particular all inscribed angles on the same arc are equal.

Proof. Draw the diameter CD through C.

Case 1: M lies on a side of the angle, say on CB. Triangle AMC is isosceles ($MA = MC$), so $\angle MAC = \angle MCA = \alpha$. The exterior angle at M is $\angle AMB = \alpha + \alpha = 2\alpha$. Done.

Case 2: M lies inside $\angle ACB$. The diameter CD splits $\angle ACB$ into $\angle ACD$ and $\angle DCB$, and $\angle AMB$ into $\angle AMD$ and $\angle DMB$. By case 1 applied twice, $\angle AMD = 2\angle ACD$ and $\angle DMB = 2\angle DCB$. Add.

Case 3: M lies outside $\angle ACB$. Say A lies between D and B as seen from C (so ray CA lies inside $\angle DCB$). Then $\angle ACB = \angle DCB - \angle DCA$ and $\angle AMB = \angle DMB - \angle DMA$. By case 1 twice, subtract. ■

Consequences.

1. Thales' theorem is the special case $\angle AMB = 180°$.
2. **Angles in the same segment are equal.** All points C on one arc see AB under the same angle.
3. **Locus.** The set of points P on one side of line AB with $\angle APB = \gamma$ is an arc of a circle through A and B (the *boog van constante hoek*). The converse direction ("if $\angle APB = \angle ACB$ then P is on the circle") is proved by showing that points inside the circle see AB under a larger angle and points outside under a smaller one (exterior angle theorem).

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4. Cyclic quadrilaterals

A quadrilateral is **cyclic** (*koordenvierhoek*) if its four vertices lie on one circle.

Theorem. In a cyclic quadrilateral $ABCD$, opposite angles add up to $180°$: $\angle A + \angle C = \angle B + \angle D = 180°$.

Proof. $\angle A$ ($= \angle DAB$) is an inscribed angle on arc BCD, and $\angle C$ ($= \angle BCD$) is an inscribed angle on the other arc BAD. The two corresponding central angles add up to $360°$, so by the inscribed angle theorem $\angle A + \angle C = \frac{1}{2} \cdot 360° = 180°$. ■

Converse. If a (convex) quadrilateral has $\angle A + \angle C = 180°$, it is cyclic.

Proof. Draw the circle through A, B, D. Points P on the arc on the other side of BD from A have $\angle BPD = 180° - \angle A = \angle C$. By the locus statement in §3, the points on that side of BD that see BD under exactly the angle $\angle C$ are exactly those on this arc. So C is on the circle. ■

Two more criteria for four points to be concyclic (both from §3):

- $\angle ACB = \angle ADB$ with C and D on the same side of AB;
- an exterior angle of the quadrilateral equals the opposite interior angle.

Example. In triangle ABC let AD and BE be altitudes (*hoogtelijnen*), meeting in H. Then $\angle ADB = \angle AEB = 90°$, so D and E lie on the Thales circle of AB: A, B, D, E **are concyclic**. Also $\angle HDC + \angle HEC = 90° + 90° = 180°$, so C, D, H, E **are concyclic**. Many olympiad problems start exactly like this (problem 12).

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5. Tangent and chord

Theorem (tangent–chord angle). Let t be the tangent at A to a circle and AB a chord. The angle between t and AB equals the inscribed angle $\angle ACB$ on the arc on the *other* side of AB.

Proof. Let M be the centre and $\angle ACB = \gamma$, so $\angle AMB = 2\gamma$ (inscribed angle). Triangle AMB is isosceles, so $\angle MAB = 90° - \gamma$. The tangent is perpendicular to MA, so the angle between t and AB is $90° - (90° - \gamma) = \gamma$. (This is the case $\gamma < 90°$; the case $\gamma \geq 90°$ is similar.) ■

You can think of the tangent–chord angle as the limit of an inscribed angle $\angle ACB$ when C slides to A: the chord CA becomes the tangent.

Power of a point (going further, very useful). If two lines through P meet a circle in A, B and in C, D, then $PA \cdot PB = PC \cdot PD$. *Proof:* triangles PAC and PDB are similar (hh, using inscribed angles). If one line is tangent at T: $PA \cdot PB = PT^2$.

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6. Analytic proof

In an **analytic proof** (*analytisch bewijs*) you put the figure in a coordinate system and compute. The art is choosing coordinates so that the computation is short.

Guidelines.

- Put a special point at the origin and a special line along an axis.
- Use symmetry: for an isosceles triangle put the base on the x-axis symmetric about the y-axis.
- Use letters for general positions: a general triangle is $A(0,0)$, $B(b,0)$, $C(c,d)$. That is fully general (any triangle can be moved and rotated into this position), so a proof with these coordinates proves the theorem for all triangles.
- Vectors are often shorter than coordinates: write $\vec{a}, \vec{b}, \vec{c}$ for position vectors.

Example 1 (Thales again). Circle $x^2 + y^2 = r^2$, $A(-r,0)$, $B(r,0)$, $C(p,q)$ on the circle. Then

$$\overrightarrow{CA} \cdot \overrightarrow{CB} = (-r-p)(r-p) + (-q)(-q) = p^2 - r^2 + q^2 = 0,$$

so $CA \perp CB$. Three lines.

Example 2 (centroid). The medians (*zwaartelijnen*) of a triangle are concurrent. With position vectors $\vec{a}, \vec{b}, \vec{c}$, the midpoint of BC is $\frac{1}{2}(\vec{b} + \vec{c})$. The point that divides the median from A in ratio $2 : 1$ is

$$\vec{a} + \tfrac{2}{3}\left(\tfrac{1}{2}(\vec{b} + \vec{c}) - \vec{a}\right) = \tfrac{1}{3}(\vec{a} + \vec{b} + \vec{c}).$$

This expression is symmetric in a, b, c, so the same point lies on all three medians: the centroid (*zwaartepunt*) $Z = \frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$.

Synthetic or analytic? Synthetic proofs explain *why*; analytic proofs are mechanical and sure to work but can get long. Wiskunde techniek asks for both. Try synthetic first; if you are stuck after ten minutes, compute.

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7. Loci

A **locus** (*meetkundige plaats*, plural *meetkundige plaatsen*) is the set of all points satisfying a condition.

Condition	Locus
$d(P,A) = r$	circle
$d(P,A) = d(P,B)$	perpendicular bisector (*middelloodlijn*) of AB
$d(P,\ell) = d(P,m)$ for intersecting lines ℓ, m	the two angle bisectors (*bissectrices*)
$\angle APB = 90°$	circle with diameter AB (minus A, B)
$d(P,A) = k \cdot d(P,B)$, $k \neq 1$	circle of Apollonius
$d(P,F) = d(P,\ell)$	parabola
$d(P,F_1) + d(P,F_2) = 2a$	ellipse
$\lvert d(P,F_1) - d(P,F_2)\rvert = 2a$	hyperbola

Method for finding a locus analytically. Call the moving point $P(x,y)$; translate the condition into an equation in x and y; simplify; recognise the curve; check whether all points of the curve really satisfy the condition (sometimes points must be removed).

Example. $A(0,0)$, $B(3,0)$, $PA = 2PB$. Squaring (allowed, both sides ≥ 0): $x^2 + y^2 = 4((x-3)^2 + y^2)$, so $3x^2 + 3y^2 - 24x + 36 = 0$, i.e. $(x-4)^2 + y^2 = 4$. A circle with centre $(4,0)$ and radius 2.

Distance from a point to a line. For the line $ax + by = c$ and point $P(p,q)$:

$$d(P,\ell) = \frac{|ap + bq - c|}{\sqrt{a^2 + b^2}}.$$

(Proof: (a,b) is a normal vector; project $\overrightarrow{QP}$ for any Q on ℓ onto it.) The same formula in 3D gives the distance from a point to a plane (§11).

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8. The parabola

The conic sections (*kegelsneden*) are the curves you get by cutting a double cone with a plane. Here we define them by distances, which is more useful.

Definition. Given a point F (the **focus**, *brandpunt*) and a line r not through F (the **directrix**, *richtlijn*), the **parabola** is the set of points P with $d(P,F) = d(P,r)$.

Equation. Take $F(0,p)$ and $r : y = -p$ (with $p > 0$). Then

$$\sqrt{x^2 + (y-p)^2} = |y+p| \iff x^2 + y^2 - 2py + p^2 = y^2 + 2py + p^2 \iff x^2 = 4py.$$

So $y = \frac{1}{4p}x^2$: every parabola $y = ax^2$ ($a > 0$) has focus $(0, \frac{1}{4a})$ and directrix $y = -\frac{1}{4a}$. The **vertex** (*top*) is midway between focus and directrix.

Reflection property. Let P be on the parabola and P' the foot of the perpendicular from P to the directrix.

Theorem. The tangent at P is the perpendicular bisector of FP'; equivalently, it bisects the angle $\angle FPP'$.

Proof. Let t be the perpendicular bisector of FP'. Since $PF = PP'$, P lies on t. Take any other point Q on t and let Q' be the foot of the perpendicular from Q to the directrix. Then $QF = QP'$ (since Q is on the bisector) and $QP' > QQ'$ (hypotenuse versus leg, as $P' \neq Q'$). So

$QF > QQ' = d(Q, r)$: Q is *outside* the parabola (on the side away from the focus). So t meets the parabola only in P and the whole parabola lies on one side of it. Hence t is the tangent at P. Triangle FPP' is isosceles, and the perpendicular bisector of the base is the bisector of the top angle. ■

Physical meaning. A ray coming in parallel to the axis (along $P'P$ extended) hits the parabola at P and reflects with equal angles to the tangent, so it passes through F. This is why satellite dishes, car headlights and solar furnaces are parabolic: all parallel rays meet in the focus.

Note this proof uses only the definition; no derivatives. You can also check it with calculus: for $x^2 = 4py$ the tangent at $P(x_0, y_0)$ has slope $\frac{x_0}{2p}$ and meets the y-axis at $(0, -y_0)$; the focus is at distance $p + y_0$ from that point, which equals PF, giving an isosceles triangle (problem 18).

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9. The ellipse and the hyperbola

Ellipse. Given foci F_1, F_2 and a number $2a > F_1F_2$, the **ellipse** is the set of points with $PF_1 + PF_2 = 2a$. (Draw it with a loop of string and two pins.)

Take $F_{1,2} = (\mp c, 0)$ with $0 \le c < a$ and put $b^2 = a^2 - c^2$. Starting from $\sqrt{(x+c)^2 + y^2} = 2a - \sqrt{(x-c)^2 + y^2}$, square, simplify to $a\sqrt{(x-c)^2 + y^2} = a^2 - cx$, square again:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

(Every step can be reversed for points with $|x| \le a$, which all points of the curve satisfy, so this equation describes exactly the ellipse.) The ellipse meets the axes in $(\pm a, 0)$ and $(0, \pm b)$: a is the semi-major axis, b the semi-minor axis, and $c^2 = a^2 - b^2$. A circle is the case $c = 0$.

Hyperbola. For $2a < F_1F_2 = 2c$, the set of points with $|PF_1 - PF_2| = 2a$ is a **hyperbola**. With $b^2 = c^2 - a^2$ the same computation gives

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

with two branches and **asymptotes** $y = \pm\frac{b}{a}x$. (For large $|x|$, $y = \pm\frac{b}{a}x\sqrt{1 - a^2/x^2} \approx \pm\frac{b}{a}x$.) The hyperbola $xy = 1$ is the case $a = b$ rotated by $45°$.

Directrix circle (*richtcirkel*). Dutch textbooks often use this equivalent definition. Let c_2 be the circle with centre F_2 and radius $2a$.

- **Ellipse:** P is on the ellipse iff $d(P, F_1) = d(P, c_2)$. For P inside c_2, the distance from P to c_2 is $2a - PF_2$, so the condition becomes $PF_1 + PF_2 = 2a$.
- **Hyperbola:** the same condition $d(P, F_1) = d(P, c_2)$, now with F_1 outside c_2, gives $|PF_1 - PF_2| = 2a$, i.e. one branch for P outside c_2 ($PF_2 - 2a = PF_1$); the other branch uses the directrix circle around F_1.

The parabola fits too: as F_2 moves to infinity, the circle c_2 becomes a straight line, the directrix. So all three conics are "equidistant from a point and a circle (or line)".

Reflection properties.

Theorem (ellipse). At a point P of the ellipse, the tangent makes equal angles with PF_1 and PF_2. That is, the tangent is the *external* angle bisector of $\angle F_1PF_2$; the normal is the internal bisector. So light from one focus reflects into the other (whispering galleries, kidney-stone lithotripters).

Proof. Let t be the external bisector at P. Reflect F_2 in t to F_2'. Because t bisects the angle between PF_1 extended and PF_2, the point F_2' lies on the ray from F_1 through P, beyond P, and $PF_2' = PF_2$. So $F_1F_2' = PF_1 + PF_2 = 2a$. For any other point Q on t: $QF_1 + QF_2 = QF_1 + QF_2' > F_1F_2' = 2a$ (triangle inequality, strict as Q is not on line F_1F_2'). So Q lies outside the ellipse. Hence t touches the ellipse only at P and the ellipse is on one side: t is the tangent. ■

Look at the proof: F_2' is at distance $2a$ from F_1, so it lies on the directrix circle around F_1. This is the same structure as the parabola proof: **the tangent at P is the perpendicular bisector of F_2 and its "mirror point" on the directrix circle.**

Theorem (hyperbola). The tangent at P is the *internal* bisector of $\angle F_1PF_2$. (Same proof with the triangle inequality reversed; problem 24 asks for the ellipse version in detail.)

Tangent line by formula. Implicit differentiation of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ gives the tangent at (x_0, y_0):

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1,$$

and similarly $\frac{x_0 x}{a^2} - \frac{y_0 y}{b^2} = 1$ for the hyperbola and $x_0 x = 2p(y + y_0)$ for $x^2 = 4py$. ("Replace one x by x_0 and one y by y_0.")

Eccentricity (going further). Each conic is also the set of points with $d(P, F) = e \cdot d(P, \ell)$ for a fixed line ℓ and number $e > 0$, the **eccentricity**: $e < 1$ ellipse, $e = 1$ parabola, $e > 1$ hyperbola. For the ellipse $e = c/a$. Planetary orbits are ellipses with the Sun at a focus (Kepler); Earth's orbit has $e \approx 0.017$.

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10. Vectors in space

Points in space have three coordinates; vectors $\vec{a} = (a_1, a_2, a_3)$ add componentwise. The **inner product** (*inproduct*, dot product) and length are

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\vec{a}|\,|\vec{b}| \cos\theta, \qquad |\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}},$$

exactly as in the plane (stage D). In particular $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$.

Cross product. The **cross product** (*uitproduct*, *vectorproduct*) of $\vec{a}$ and $\vec{b}$ is

$$\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1).$$

Memory aid: write $\vec{a}$ above $\vec{b}$; for the first coordinate cover the first column and take "up-left times down-right minus up-right times down-left" of the 2×2 block that remains; for the second component use the cyclic order (third, first).

Properties.

1. $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$. *Check:* $\vec{a} \cdot (\vec{a} \times \vec{b}) = a_1(a_2b_3 - a_3b_2) + a_2(a_3b_1 - a_1b_3) + a_3(a_1b_2 - a_2b_1) = 0$, since all six terms cancel in pairs.
2. $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$ and $\vec{a} \times \vec{a} = \vec{0}$.
3. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$, which is the **area of the parallelogram** spanned by $\vec{a}$ and $\vec{b}$. This follows from the Lagrange identity $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$ (problem 26).
4. The direction follows the **right-hand rule**: $(1, 0, 0) \times (0, 1, 0) = (0, 0, 1)$.
5. $\vec{a} \times \vec{b} = \vec{0}$ iff $\vec{a}$ and $\vec{b}$ are parallel (or one is zero).

Example. $(1,2,3) \times (4,5,6) = (2\cdot 6 - 3\cdot 5,\ 3\cdot 4 - 1\cdot 6,\ 1\cdot 5 - 2\cdot 4) = (-3,6,-3)$. Check: $(-3,6,-3)\cdot(1,2,3) = -3+12-9 = 0$. ✓

Volume. The parallelepiped spanned by $\vec{a},\vec{b},\vec{c}$ has volume $|(\vec{a}\times\vec{b})\cdot\vec{c}|$ (base area times height). A tetrahedron has one sixth of that.

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11. Lines and planes in space

Line. $\vec{x} = \vec{p} + \lambda\vec{r}$: a support vector (*steunvector*) $\vec{p}$ and a direction vector (*richtingsvector*) $\vec{r}$. In space a line has no single equation like $ax+by=c$; you need the vector form (or two equations).

Plane. A plane is determined by a point $\vec{p}$ and a **normal vector** (*normaalvector*) $\vec{n} \neq \vec{0}$ perpendicular to it. Point X lies in the plane iff $\overrightarrow{PX} \perp \vec{n}$:

$$\vec{n}\cdot(\vec{x}-\vec{p}) = 0 \iff n_1x + n_2y + n_3z = d \quad \text{with } d = \vec{n}\cdot\vec{p}.$$

So **the coefficients of x,y,z in the equation of a plane form a normal vector**.

Plane through three points. $A(1,0,0)$, $B(0,2,0)$, $C(0,0,3)$: $\overrightarrow{AB} = (-1,2,0)$, $\overrightarrow{AC} = (-1,0,3)$, normal $\vec{n} = \overrightarrow{AB}\times\overrightarrow{AC} = (6,3,2)$. Equation $6x+3y+2z=6$ (fill in A). Check with B and C: $6=6$, $6=6$. ✓ (This plane has intercepts $1,2,3$, and indeed $\frac{x}{1}+\frac{y}{2}+\frac{z}{3}=1$.)

Angles.

- Between two planes: the angle between their normal vectors (take the acute one): $\cos\theta = \dfrac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$.
- Between a line (direction $\vec{r}$) and a plane (normal $\vec{n}$): $\sin\theta = \dfrac{|\vec{r}\cdot\vec{n}|}{|\vec{r}||\vec{n}|}$ (the complement of the angle with the normal).

Distance from a point to a plane. For $V: ax+by+cz=d$ and $P(p,q,s)$:

$$d(P,V) = \frac{|ap+bq+cs-d|}{\sqrt{a^2+b^2+c^2}}.$$

Intersection of two planes is a line with direction $\vec{n}_1\times\vec{n}_2$ (perpendicular to both normals); find one point by setting one coordinate to 0 (problem 22).

Intersection of a line and a plane. Substitute $\vec{x} = \vec{p}+\lambda\vec{r}$ into the plane equation and solve for λ.

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12. Common mistakes

1. **Proving from the figure.** "It looks perpendicular" is not a reason. Each step needs a named fact.
2. **Assuming what you want to prove.** In particular, using the converse of a theorem instead of the theorem.
3. **Forgetting the case distinction** in the inscribed angle theorem, or assuming the centre is inside the triangle.
4. **Using "opposite angles add to $180°$" for a quadrilateral not known to be cyclic.**
5. **Mixing up $c^2 = a^2 - b^2$ (ellipse) and $c^2 = a^2 + b^2$ (hyperbola).**

6. **Squaring without care** when deriving a locus: squaring can add points (check them), and $\sqrt{(\ldots)} = |y + p|$ needs the absolute value.
7. **Cross product sign errors**, especially in the middle component. Always check perpendicularity with the dot product afterwards.
8. **Taking the obtuse angle** between planes or lines when the acute one is asked for (use $|\cdot|$ in the formula).
9. **Thinking a line in space has an equation** $ax + by + cz = d$**.** That is a plane.

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13. Going further

- **Ptolemy's theorem.** In a cyclic quadrilateral $ABCD$: $AC \cdot BD = AB \cdot CD + AD \cdot BC$ (problem 25). Applied to a rectangle it gives Pythagoras; applied to a regular pentagon it gives the golden ratio.
- **The nine-point circle.** The midpoints of the sides, the feet of the altitudes and the midpoints of AH, BH, CH all lie on one circle. Prove it using Thales circles.
- **Dandelin spheres.** Why is the cross-section of a cone with a plane an ellipse with the foci defined as above? Put two spheres in the cone touching the plane: they touch it exactly in the foci. Beautiful and short; look it up after you have proved the reflection property.
- **Projective geometry.** Pascal's theorem: for six points on a conic, the three intersection points of opposite sides of the hexagon are collinear.
- **Olympiad geometry.** The olympiad module O4 builds on everything in §1–5: power of a point, radical axes, homothety.
- **University connection.** Cross products and planes return in linear algebra (determinants, the parent's module 1.3) and in multivariable calculus (module 1.5).